

Introduction to Gene Finding

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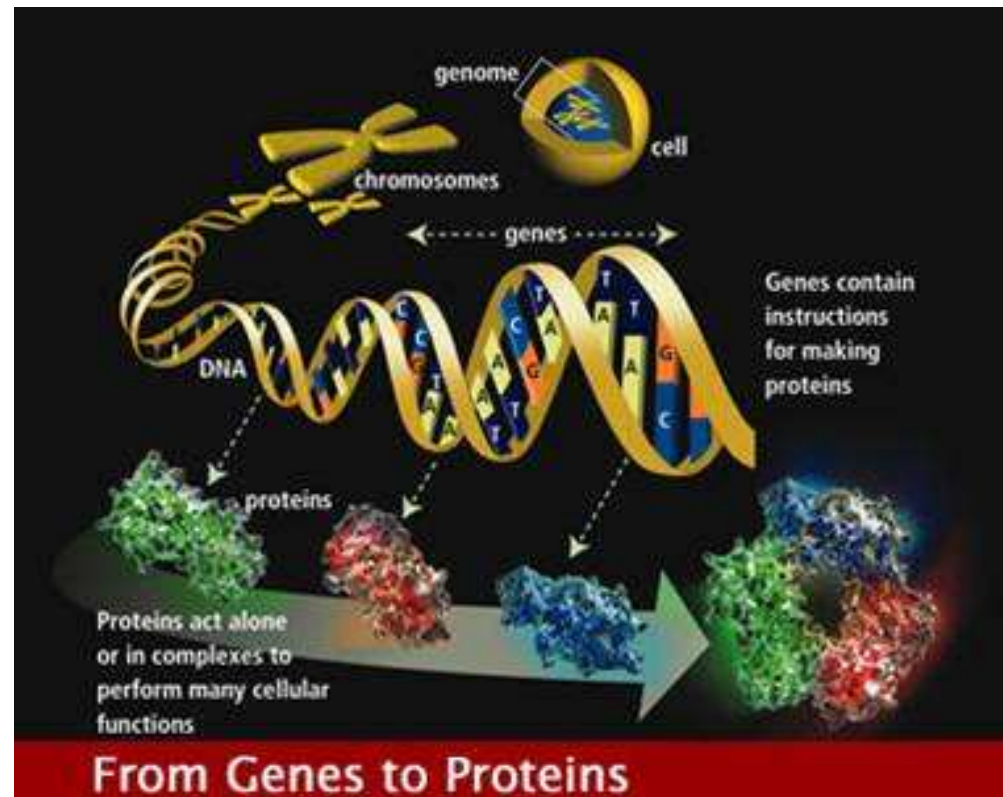
Date: Aug. 27, 2004

Outline

- The Problem
- Homology Method
- Ab Initio Method
- The K_a/K_s Ratio Test for Accessing the Protein-Coding Potential of Genomic Regions

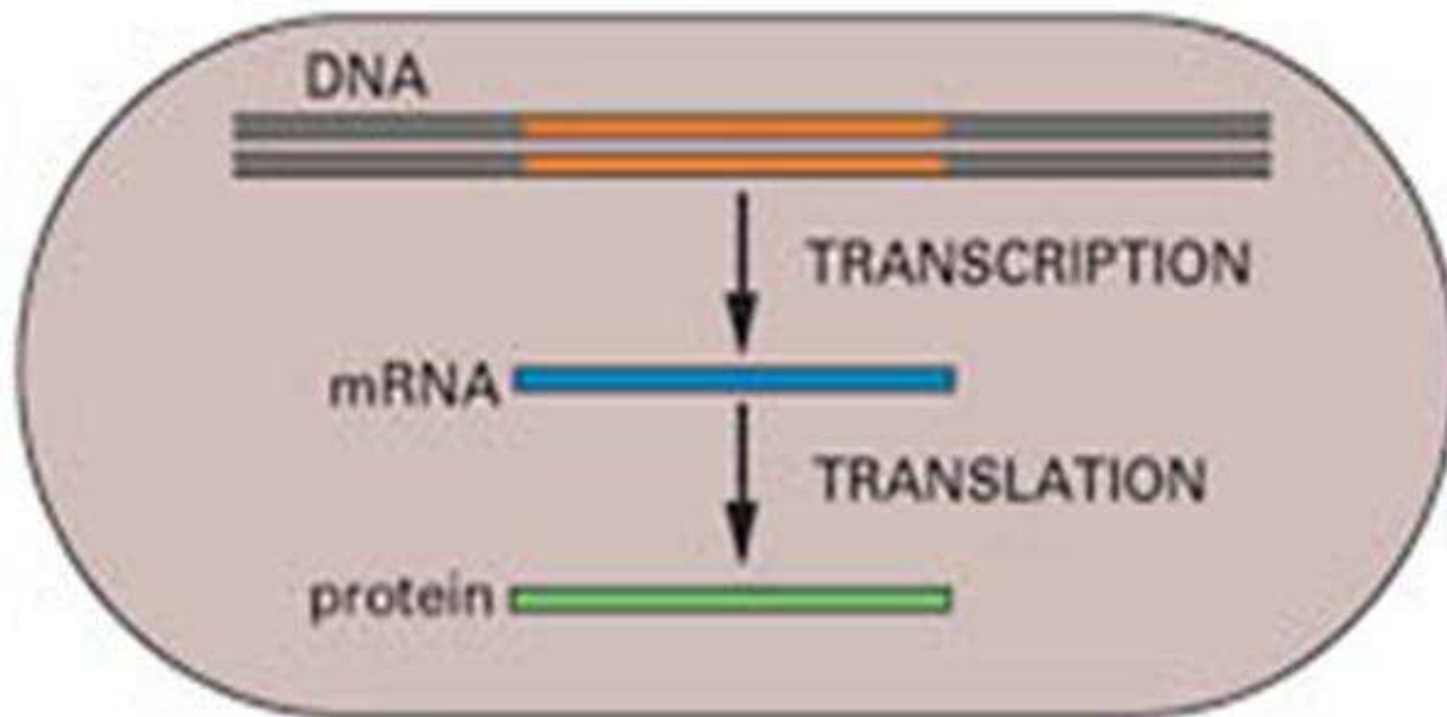
The Problem

- Gene finding is about detecting coding regions and infer gene structure.



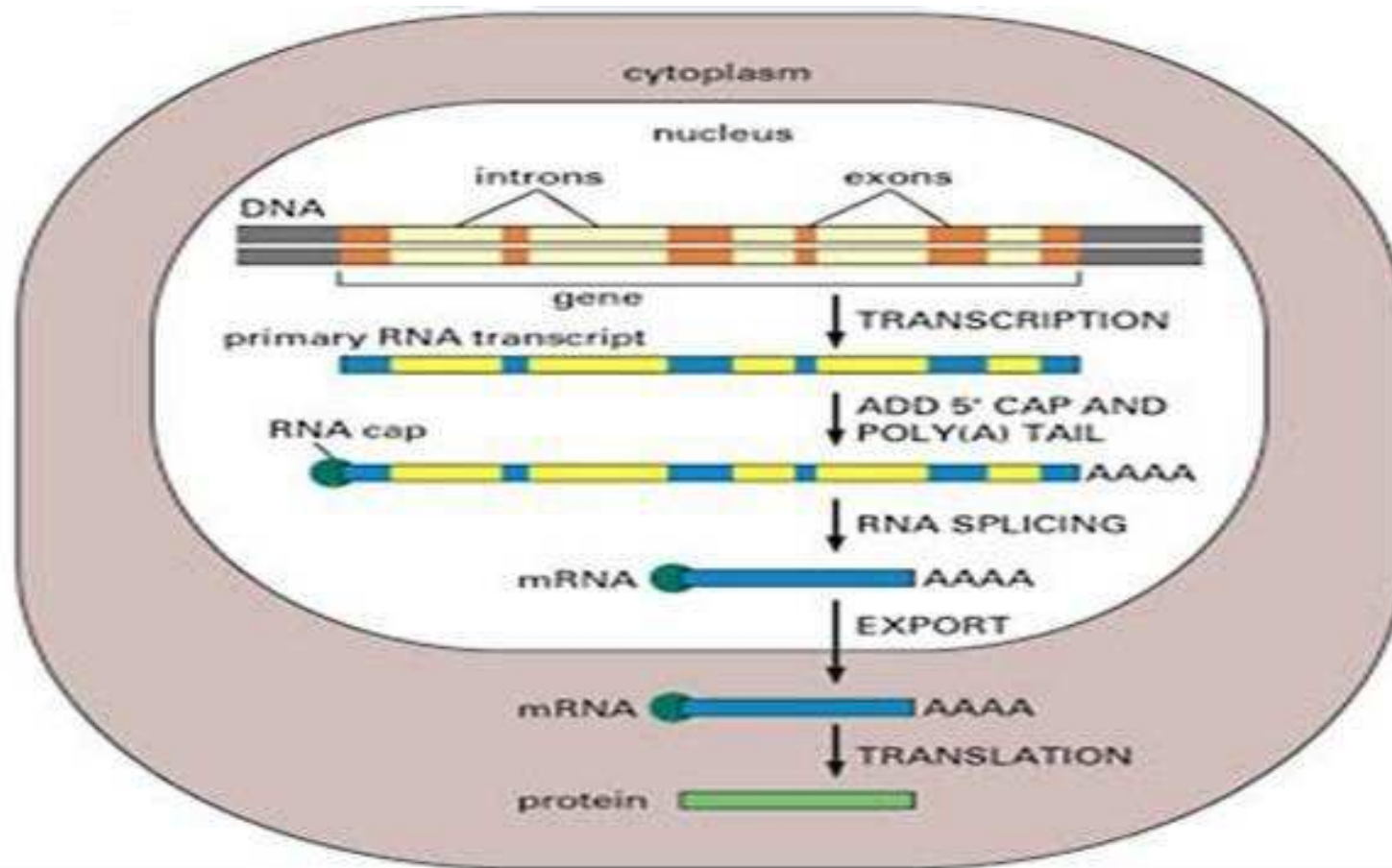
Prokaryotes

- High gene density and simple gene structure.
- Short genes have little information.
- Overlapping genes.

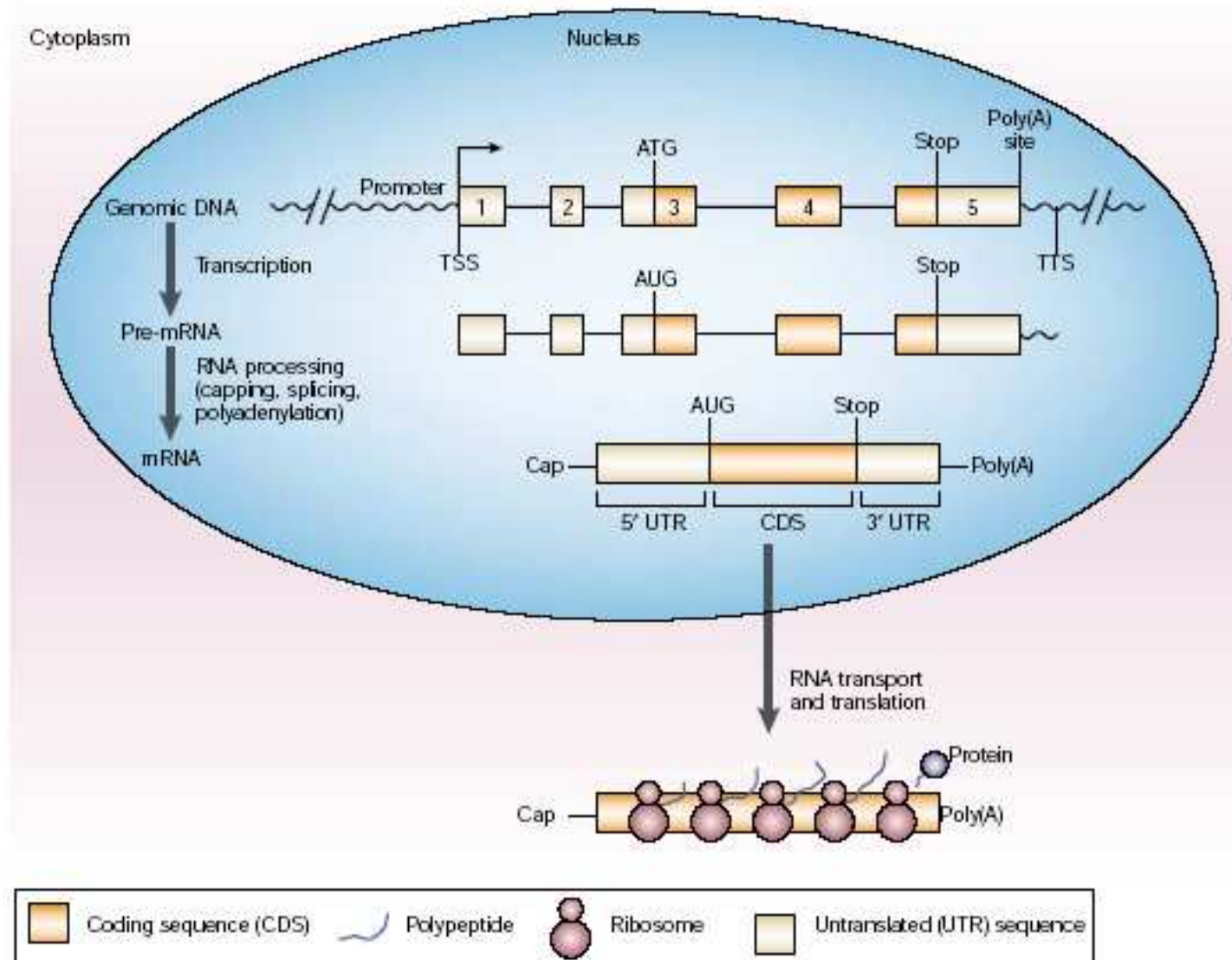


Eukaryotes

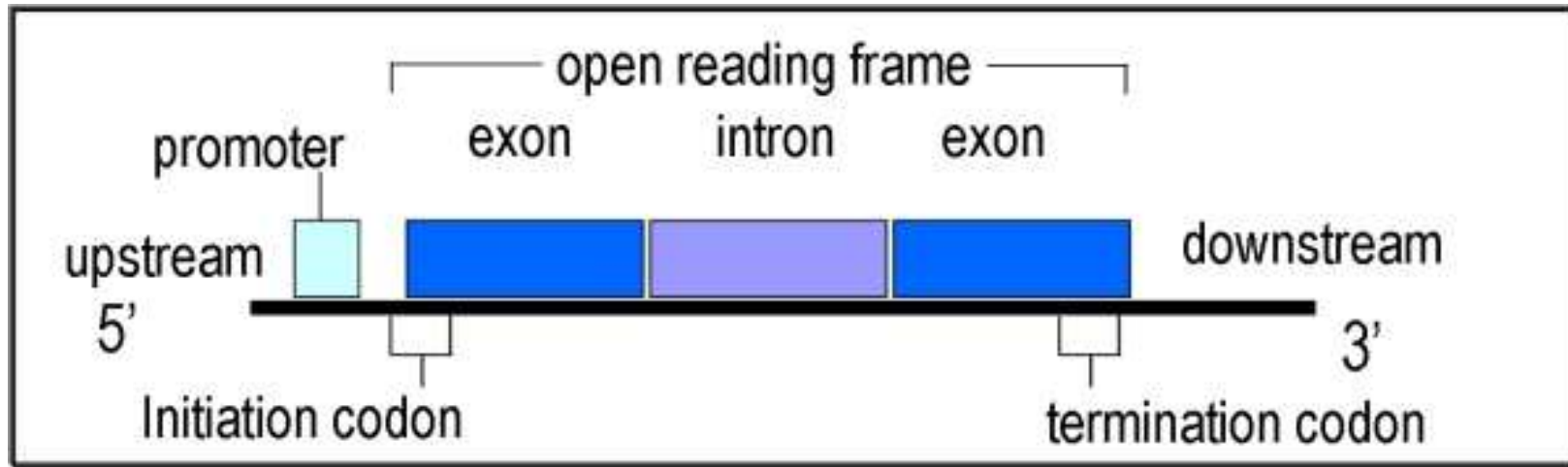
- Low gene density and complex gene structure.
- Alternative splicing.
- Pseudo-genes.



Eukaryotic Gene Expression

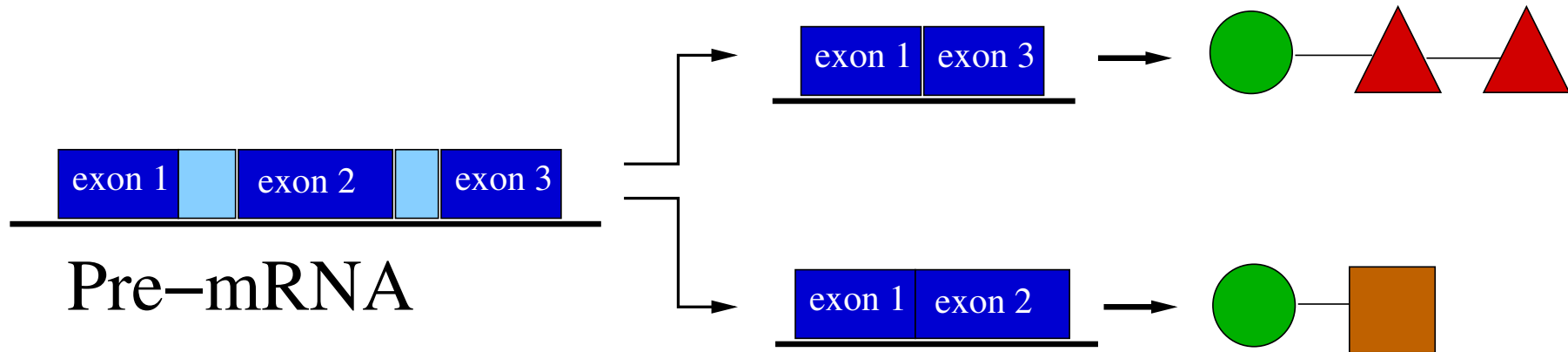


Eukaryotes Genome Structure



- Promoter: nucleotide sequence to which RNA polymerase binds to initiate transcription
- Initiation codon: usually but not exclusively ATG
- Termination codon: TAA, TAG, TGA

Alternative Splicing



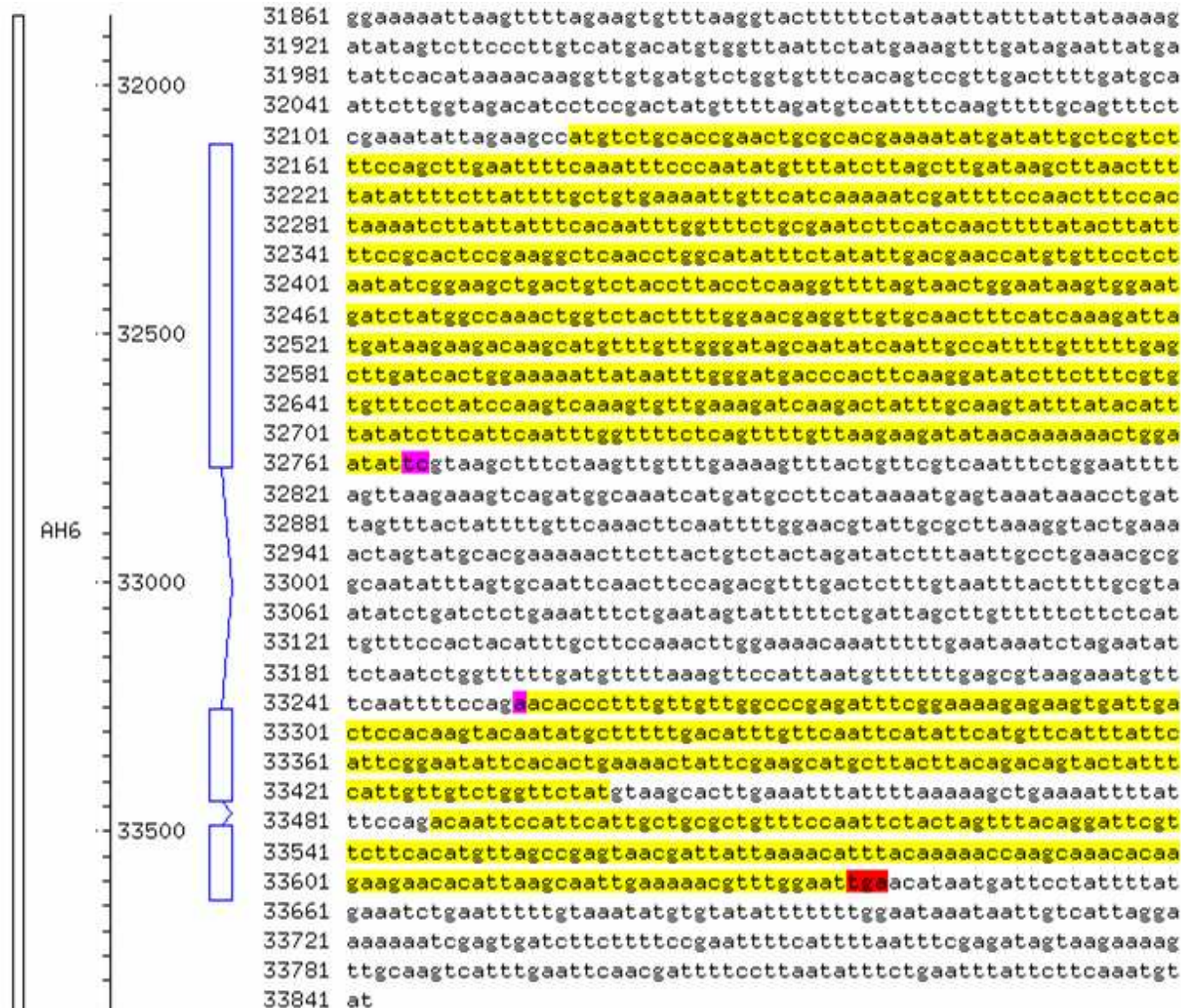
- Same gene has multiple transcripts.
- Depends on exons incorporated into mature mRNA.
- Average human gene has 10-15 exons, encodes ≥ 3 proteins.

Finding Genes in Genomic DNA

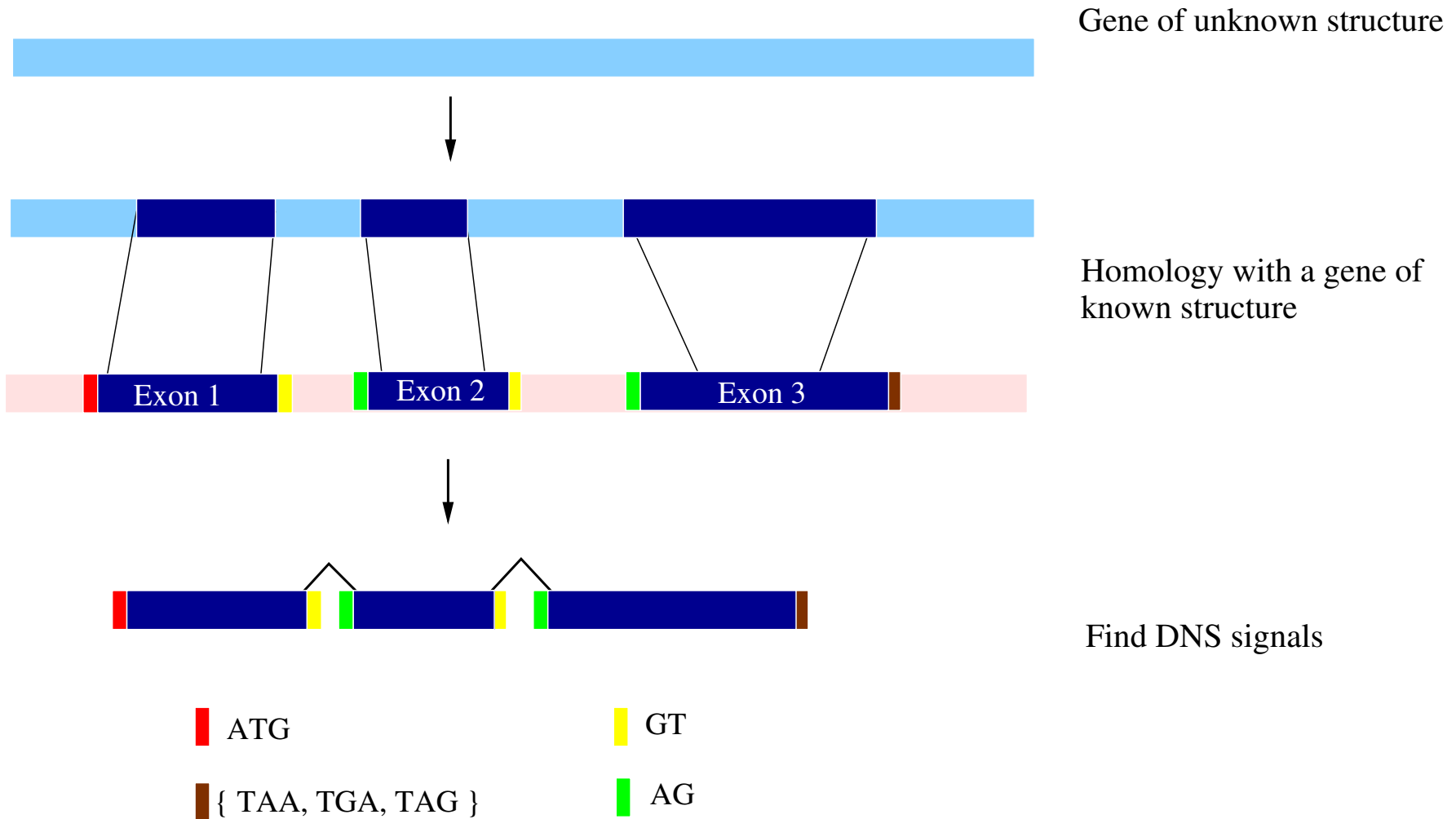
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AGAGCCTCTTCATTCTACACATGTTTCGTAAAGTTTTCTATTTTTGAATGATGTGAATTTACTACCTATTGAACAAT
TTTAAAACTTTATGCATGTGTTTGAAGGTACATTATGATTTTTCTGGAAGAATATACCTTGTTGGTGGCAGTTGC
CCCCGGCAGTTGGGAGTGCAGAGGAGAAAAGGACTTTTACTTGCCATTTACACACTTTGAGATCATTTCATTGT
AACCATGTACATTAGGACTTCTCAAGTACAAATGACTAATTATAATGACTTATGTTTATAATGACTTATGTTGAC
ATATTTCTAAATATGTAAGAAACACGTAGGTATGTATTTCTACGTATTATCTGGGTTTCCTTTGAGCAGTGGGAT
```

- Which regions codes for a protein?
- Where does the gene start and end?
- Where are the exon-intron boundaries?

Example



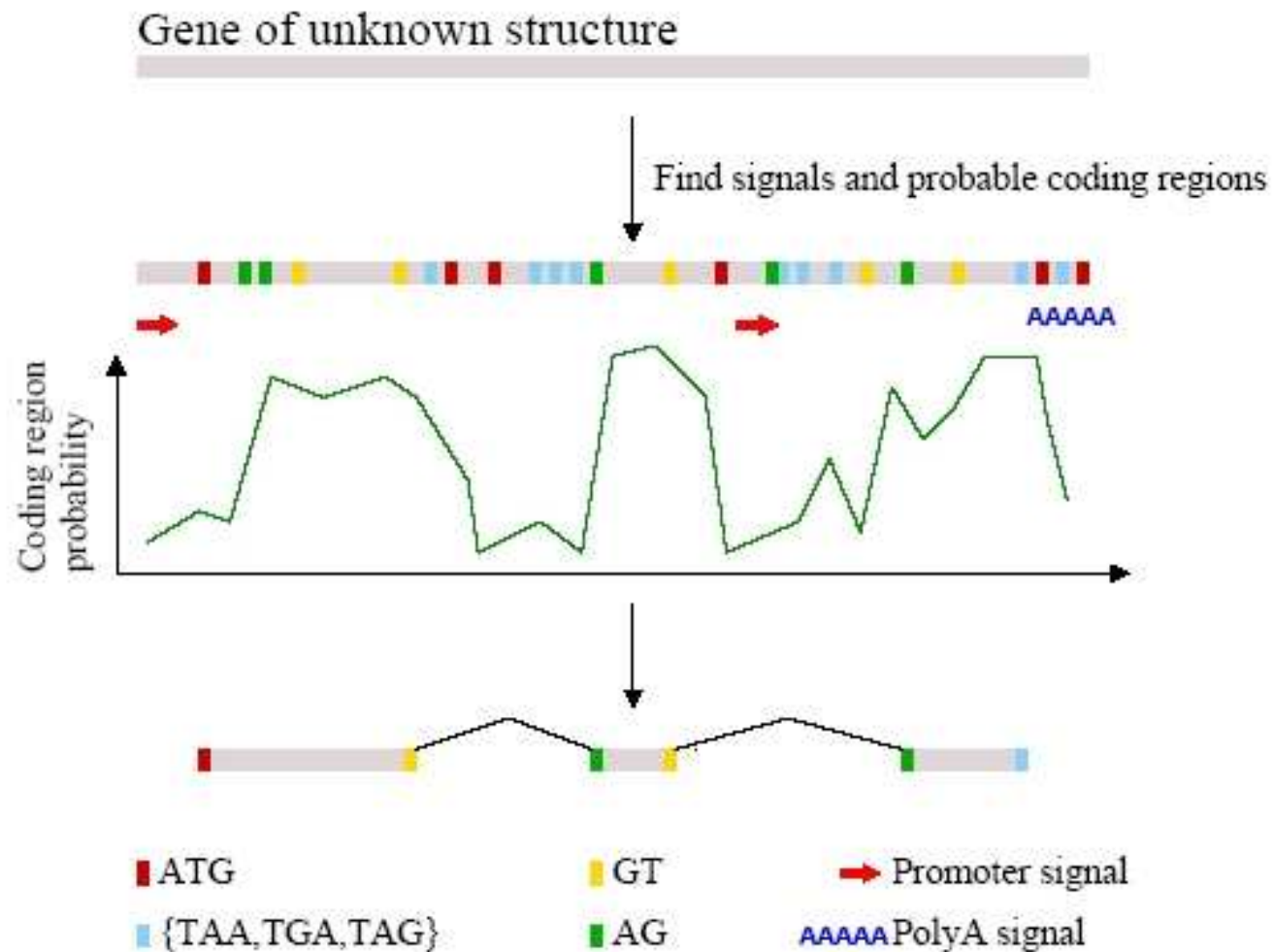
Homology Method (A Simple View)



Ab Initio Method

- Integration of *signal detection* and *coding statistics*.
- *Signal detection* and *coding statistics* are deduced from a training set.
- Probabilistic frameworks are used to infer a probable gene structure.
- A solid scoring system can be used to evaluate the predictions.

Ab Initio Method (contd.)



Signal Detection

- Detection short DNA motifs (promoters, start/stop codons, splice sites, ...).
- A number of methods are used for signal detection (Pattern recognition, Weight matrices, ...).

Coding Statistics

- Unequal usage of codons in coding regions is universal feature of genomes.
- Use coding statistics to differentiate between coding and non-coding regions.

		2nd base in codon				
		U	C	A	G	
1st base in codon	U	Phe Phe Leu Leu	Ser Ser Ser Ser	Tyr Tyr STOP STOP	Cys Cys STOP Trp	3rd base in codon
	C	Leu Leu Leu Leu	Pro Pro Pro Pro	His His Gln Gln	Arg Arg Arg Arg	
	A	Ile Ile Ile Met	Thr Thr Thr Thr	Asn Asn Lys Lys	Ser Ser Arg Arg	
	G	Val Val Val Val	Ala Ala Ala Ala	Asp Asp Glu Glu	Gly Gly Gly Gly	

Coding Statistics (contd.)

Homo sapiens [gbpri]: 66017 CDS's (28570206 codons)

fields: [triplet] [frequency: **per thousand**] ([number])

UUU 17.2(490400)	UCU 14.9(426242)	UAU 12.1(345438)	UGU 10.3(293297)
UUC 20.3(581388)	UCC 17.6(503254)	UAC 15.4(439821)	UGC 12.5(356544)
UUA 7.5(213016)	UCA 12.0(343640)	UAA 0.7(19732)	UGA 1.2(35277)
UUG 12.8(364839)	UCG 4.5(127858)	UAG 0.6(16300)	UGG 13.1(372973)
CUU 13.0(371650)	CCU 17.4(498122)	CAU 10.7(305376)	CGU 4.6(132373)
CUC 19.6(561298)	CCC 20.0(572502)	CAC 15.0(428879)	CGC 10.7(304756)
CUA 7.1(201655)	CCA 16.9(481847)	CAA 12.1(344309)	CGA 6.3(178602)
CUG 40.1(1145864)	CCG 7.0(201372)	CAG 34.4(982889)	CGG 11.7(333010)
AUU 15.8(451325)	ACU 13.0(372019)	AAU 16.7(478164)	AGU 12.1(345029)
AUC 21.0(600844)	ACC 19.1(545309)	AAC 19.2(547455)	AGC 19.4(555386)
AUA 7.3(208121)	ACA 14.9(426831)	AAA 24.1(687139)	AGA 11.7(335217)
AUG 22.2(633191)	ACG 6.2(175794)	AAG 32.2(918884)	AGG 11.7(333960)
GUU 11.0(313959)	GCU 18.6(531029)	GAU 22.0(628756)	GGU 10.8(308468)
GUC 14.6(417313)	GCC 28.3(807553)	GAC 25.5(728516)	GGC 22.5(643650)
GUA 7.1(202125)	GCA 16.0(457636)	GAA 29.0(827714)	GGA 16.4(469514)
GUG 28.5(815668)	GCG 7.6(216083)	GAG 40.0(1142235)	GGG 16.5(470796)

Comparative Genomics

- Problems of current gene prediction tools.
 - High false-positive and false-negative.
 - Decreased accuracy toward long exons and single exon genes.
- Current gene prediction tools use
 - A similar set of criteria.
 - Try to infer all necessary information from one sequence.
- Gene prediction algorithms should make extensive use of cross-species comparison.

Cross-Species Comparison

- Based on two key assumptions.
- First, nondistantly-related species, such as human and mouse, share the vast majority of their genes. In fact, it has been estimated that only 1% of human genes have no homology with other animals.
- Second, most genes are subject to purifying selection with stronger selective constraints for nonsynonymous changes than for synonymous ones (Li 1997; Makalowski and Boguski 1998).

The Ka / Ks Ratio Test

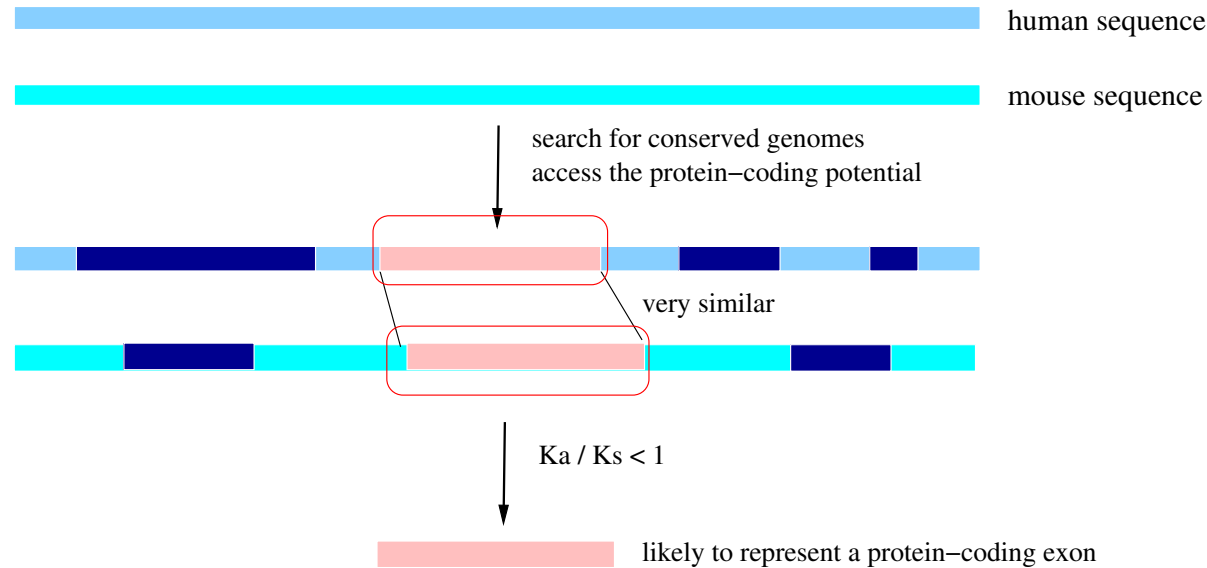
- Nucleotide substitutions in protein-coding regions are divided into two classes.
 - Change amino acid (nonsynonymous).
 - Do not (synonymous).

		2nd base in codon				
		U	C	A	G	
1st base in codon	U	Phe Phe Leu Leu	Ser Ser Ser Ser	Tyr Tyr STOP STOP	Cys Cys STOP Trp	U C A G
	C	Leu Leu Leu Leu	Pro Pro Pro Pro	His His Gln Gln	Arg Arg Arg Arg	U C A G
	A	Ile Ile Ile Met	Thr Thr Thr Thr	Asn Asn Lys Lys	Ser Ser Arg Arg	U C A G
	G	Val Val Val Val	Ala Ala Ala Ala	Asp Asp Glu Glu	Gly Gly Gly Gly	U C A G
						3rd base in codon

The K_a / K_s Ratio Test (contd.)

- Two normalized values.
 - K_a , the number of nonsynonymous substitutions per nonsynonymous site.
 - K_s , the number of synonymous substitutions per synonymous site.
- On average, $K_a < K_s$, the ratio K_a/K_s has been found to be significantly smaller than one for most protein-coding regions.

Method



- If a genomic region is very similar between two species and contains a reading frame that satisfies the requirement $K_a / K_s < 1$, then this region is likely to represent a protein-coding exon.

Result (1)

Table 1. Descriptive Statistics of Human Protein-Coding Exons

Exons	No. of exons	Length ^a			GC%		% Identity with mouse
		range	mean	median	overall	third pos.	
All ^c	1219	30–1842	147	123	52.1 ± 9.6	59.4 ± 19.3	86.5 ± 8.0
Initial ^d	114	30–1842	222	190	59.6 ± 10.2	72.7 ± 17.0	
Internal	991	30–624	132	123	51.0 ± 9.1	57.5 ± 19.1	
Terminal ^d	114	30–1167	203	190	54.5 ± 10.2	63.9 ± 18.7	
Single-exon genes	25	327–4737	1376	1228	56.1 ± 9.6	68.2 ± 19.0	

^aComparisons of human and mouse sequences often involve insertion/deletion events. Therefore, the alignable length of exons in this study is sometimes smaller than their actual length. However, in the majority of cases (1104 of 1244, or 88%) alignable length is equal to the actual lengths of human and mouse sequences.

^bMinimal alignable length was set to 30 bp.

^cExcluding single-exon genes.

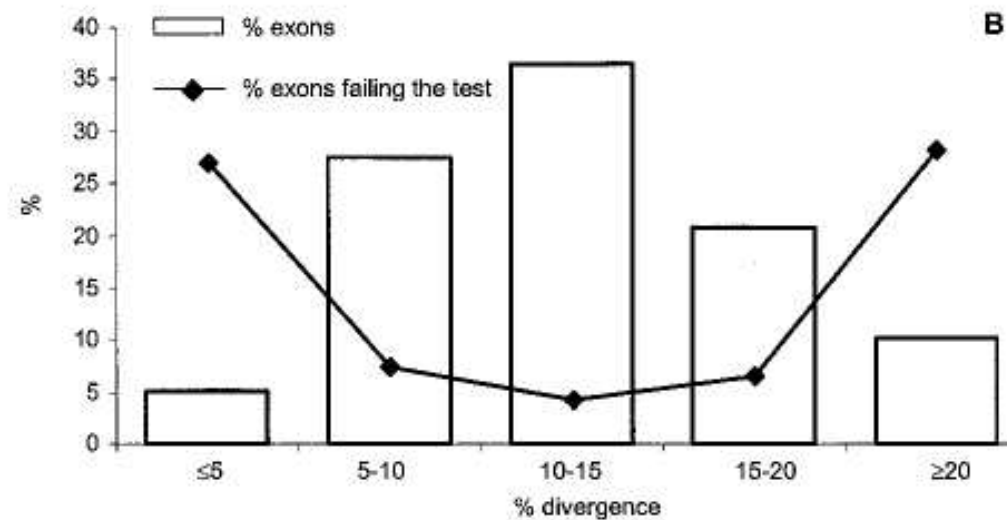
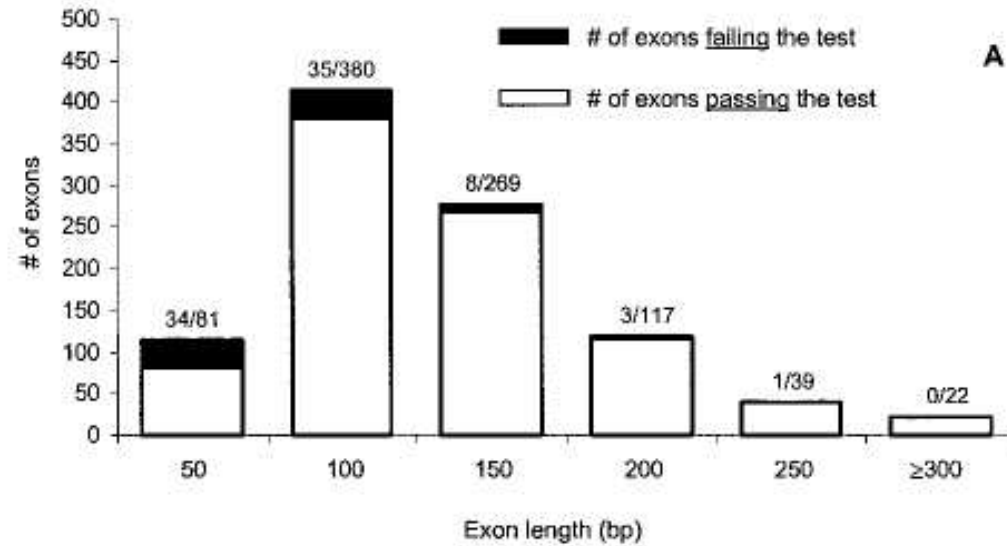
^dOnly coding parts of the initial and terminal exons were included.

Result (2)

Table 2. Summary Statistics for K_a and K_s

Exons	K_a			K_s			K_a/K_s		
	mean (SD)	median	range	mean (SD)	median	range	mean(SD)	median	range
First	0.083 (0.111)	0.041	0–0.615	0.920 (1.134)	0.620	0–9.802	0.139 (0.237)	0.057	0.001–2.024
Internal	0.062 (0.125)	0.034	0–1.863	0.820 (0.812)	0.602	0–7.904	0.100 (0.150)	0.054	0.001–2.138
Last	0.082 (0.152)	0.045	0–1.340	0.869 (0.747)	0.693	0–4.053	0.122 (0.207)	0.061	0.001–1.600
Single-exon	0.055 (0.051)	0.045	0–0.209	0.665 (0.273)	0.646	0–1.335	0.088 (0.080)	0.071	0.005–0.383
All	0.065 (0.125)	0.036	0–0.863	0.830 (0.835)	0.608	0–9.802	0.105 (0.164)	0.054	0.001–2.138

Result (3)



Computer Simulation

Table 3. Proportion of Random Sequences with K_A/K_S Significantly Smaller than 1 ($P < 0.05$)^a

% Divergence	Length (codons/nucleotides)					
	17/5	34/102	50/150	67/201	100/300	334/1002
5	3.0 (2.0–4.1)	2.2 (1.3–3.1)	2.5 (1.5–3.5)	2.5 (1.5–3.5)	2.0 (1.1–2.9)	2.2 (1.3–3.1)
10	2.9 (1.9–3.9)	2.8 (1.8–3.8)	2.6 (1.6–3.6)	1.8 (1.0–2.6)	2.2 (1.3–3.1)	1.9 (1.1–2.7)
15	5.0 (3.6–6.4)	3.3 (2.2–4.4)	1.7 (0.9–2.5)	2.1 (1.2–3.0)	3.5 (2.4–4.6)	1.9 (1.1–2.7)
20	4.8 (3.5–6.1)	2.9 (1.5–3.5)	2.9 (1.9–3.9)	2.5 (1.5–3.5)	2.5 (1.5–3.5)	1.1 (0.5–1.7)

^aThe values in parentheses indicate the 95% confidence interval.

Thanks

Thanks